Rich Chau

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RESEARCH INTERESTS

Applied Mathematics & Numerical Analysis, Video Compression & Digital Signal Processing, Systems Programming (C/C++), Screen Content Coding (SCC), High-Throughput Stream Decoders, Rate-Distortion Optimization.

EDUCATION

University of California, Irvine *Irvine, CA*
Degree in Applied Mathematics 2026

- **Academic Research / Thesis:** *TimeFlowCodec: An Orthogonal Video Coding Architecture for Screen and Synthetic Content via Per-Pixel Temporal Regression* (55-page monograph).
- **Focus Areas:** Applied Linear Algebra, Numerical Optimization, Signal Processing, High-Performance Systems.

PUBLICATIONS & MANUSCRIPTS

- **R. Chau**, "TimeFlowCodec: An Orthogonal Video Coding Paradigm for Screen and Synthetic Content via Per-Pixel Temporal Regression," *IEEE Conference on Multimedia / Data Compression (Under Review / arXiv Preprint)*, 5 pages, 2026.
- **R. Chau**, "Per-Pixel Vectorization for Temporal Video Compression," *Proceedings of Applied Mathematics in Multimedia and Signal Processing*, 2023.
- **R. Chau**, "TimeFlowCodec Bitstream Specification (Version 2.0 Draft)," *Alliance for Open Media (AOMedia) / IETF Internet-Draft Technical Contribution*, 2026.

RESEARCH SYSTEMS & OPEN-SOURCE ENGINEERING

TimeFlowCodec (TFC) · *Lead Architect & Creator* github.com/rchau8502/TimeFlowCodec
An orthogonal, patent-free video coding standard engineered for screen, terminal, and slide content.

- Formulated a novel video coding paradigm that discards 2D spatial DCT block transforms in favor of 1D continuous per-pixel Ordinary Least Squares (OLS) temporal regression with native 4:4:4 RGB representation.
- Eliminated high-frequency Gibbs ringing artifacts ($> 10\%$ overshoot) and chroma bleeding around text edges.
- Achieved an average -26.8% **BD-Rate reduction** over H.264 (up to -41.2% on slide decks) evaluated under standard ITU-T VCEG-M33 Common Test Conditions on JVET Screen Content Coding sequences.
- Designed a Leaky-Bucket Rate Controller (CRF 0–63, CBR, VBR) and Lagrangian Rate-Distortion Optimization ($J = D + \lambda R$).

libtfc: Standalone ANSI C Reference Decoder · *Lead Developer* 2026
Ultra-lightweight, zero-dependency C99 decoder engine.

- Engineered an ANSI C99 library and standalone CLI player with zero runtime dependencies.
- Delivered $>$ 840 **FPS** decode throughput ($>$ 2,200 MB/s raw RGB24 at 720p) and $>$ 384 FPS at 1080p, utilizing an $\mathcal{O}(1)$ sequential line-buffer without requiring multi-frame Decoded Picture Buffers (DPB).
- Demonstrated a 75% **reduction in package power dissipation** (0.85 W vs. 3.42 W) over software AV1 via macOS `powermetrics`.

Privacy-First Asynchronous Research Testbed · *Full-Stack Architect* 2026
Real-world data collection and offloading framework for computational video research.

- Architected a zero-PII Telegram frontend with salted HMAC-SHA256 pseudonymization and student email verification.
- Built an event-driven serverless ingestion API on Vercel backed by PostgreSQL/Supabase with strict Row Level Security (RLS) and atomic `FOR UPDATE SKIP LOCKED` job queueing.
- Implemented an outbound-only polling worker on university Slurm HPC clusters executing automated batch compression.

TECHNICAL SKILLS

- **Languages:** C (ANSI C99), C++, Python, SQL, Bash, x86/ARM Assembly, LaTeX, HTML/CSS.
- **Video & Signal Processing:** Video Coding Standards (H.264/AVC, H.265/HEVC, AV1, VVC), ITU-T VCEG-M33 Bjontegaard Delta (BD-Rate), FFmpeg, libavcodec, Discrete Cosine Transform (DCT), OLS Temporal Regression, Wavelets.
- **Systems & Architecture:** Slurm Cluster Scheduler, Linux/POSIX Kernel APIs, Git, Zstandard, SMP Multi-Threading, SIMD Vectorization, Memory-Mapped I/O (`mmap`), PySide6/Qt GUI.

RELEVANT COURSEWORK

- **Computer Science:** Advanced Operating Systems, Analysis of Algorithms, Computer Systems Architecture, Distributed Computing, Machine Learning, Data Structures & Algorithms.
- **Mathematics:** Linear Algebra & Matrix Analysis, Numerical Analysis, Continuous Optimization, Discrete Mathematics, Probability & Stochastic Processes.

PRESENTATIONS & ACADEMIC SERVICE

- *TimeFlowCodec Defense Presentation Deck:* 20-slide LaTeX Beamer deck and interactive HTML5 inspection suite with real-time Gibbs overshoot raster loupe and committee Q&A guide (2026).
- *Open-Source Mentorship & Review:* Active code maintainer and reviewer for research repositories, reproducible video coding benchmark harnesses, and student testbeds.